

The Algebraic Bridge: FFGFT and $\mathbb{Z}_3\mathbb{C}$

Frobenius Automorphism, Charge Classification and
Five Structural Questions between $T^4/\mathbb{Z}_3$ and $\text{GF}(9)$

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The Algebraic Bridge: FFGFT and $\mathbb{Z}_3\mathbb{C}$

Abstract

Two independently developed frameworks — the Fundamental Fractal Geometric Field Theory (FFGFT) with its compact $T^4/\mathbb{Z}_3$ geometry and Matzke's GALG with ternary algebra $\mathbb{Z}_3\mathbb{C}$ — arrive at the same $\mathbb{Z}_3$ structure by entirely different routes. This document shows algebraically *why* this is not a coincidence, and derives new physical relations from it.

The key is the *Frobenius automorphism* $x \mapsto x^3$ in GF(9): it simultaneously realises the FFGFT sector pairing $k \leftrightarrow -k$, Matzke's vacuum conjugation $V_s = V^\dagger$, and the field conjugation in $\mathbb{Z}_3\mathbb{C}$ **[B]**.

The main new results:

- The Wolfenstein parameter of the CKM matrix satisfies $\lambda = \xi^{1/6}$ (0.5% deviation) **[K]**; the exponent $1/6 = 1/(2 \cdot 3)$ is a motivated stipulation: 6 is the degree of the compositum $\text{GF}(729) = \text{GF}(9) \cdot \text{GF}(27)$ over GF(3), since $\text{GF}(9) \not\subset \text{GF}(27)$.
- The CKM element $|V_{ub}| = \sqrt{\xi}/3 = \sqrt{\xi} \cdot (C_2 - 1)$ (0.8% deviation), where $C_2 = 4/3$ is the Casimir value of SU(3) **[K]**.
- GF(9) and GF(27) classify the particle spectrum topologically — but exact mass values require $\xi \in \mathbb{R}$ **[B]**.

Notation: Symmetric $\mathbb{Z}_3\mathbb{C}$ after Matzke: $a + bi$ with $a, b \in \{-1, 0, +1\}$, $+1 + 1 = -1$ (ternary inline carry).

Guide: what this document shows

The basic problem. The three quark generations mix with strongly different strengths. The Wolfenstein parameter $\lambda \approx 0.225$ describes this hierarchy empirically — but why exactly this value? Why three generations? Why do the muon and electron have such different masses, even though both are leptons?

What GF(9) and GF(27) contribute. The finite field GF(9) has exactly 9 elements. Its Frobenius automorphism $x \mapsto x^3$ partitions them into three classes: fixed points $\{0, +1, -1\}$ (the values of the triality Δw), and three conjugate pairs $(k, -k)$ (the sector pairing). GF(27) extends this: it has degree 3 over GF(3) (GF(9) is not a subfield of GF(27); degree 6 is reached only by the compositum GF(729)), and the 8 three-cycles correspond to 8 quark-colour states in 3 generations.

What ξ does that GF(9) cannot. GF(9) is a finite field — it has exactly 9 elements, and none of them equals $m_e/m_p \approx 4 \times 10^{-23}$. Exact particle masses cannot be represented in a finite field. The transition to physical scales occurs through $\xi = 4/30\,000$, whose denominator 3 forms a singularity in GF(3) — this is algebraically the transition from discrete topology to continuous physics.

The surprising part. Despite this, the combination of the field degrees of GF(9), GF(27) and ξ yields precise numerical predictions for CKM parameters that were previously known only empirically. This suggests that the Wolfenstein hierarchy is not a free parametrisation but follows from algebraic structure.

Notation and symbols

Symbol	Meaning	Source
<i>Status markers</i>		
[B]	algebraically proved, verification script available	Doc. 190
[K]	derivable from ξ (internally consistent)	Doc. 190
[S]	open bridge (sketch, no proof yet)	Doc. 190
<i>Fields and algebras</i>		
GF(p^n)	Finite field with p^n elements	Galois theory
$\mathbb{Z}_3\mathbb{C}$	Sym. GF(9): $a + bi$, $a, b \in \{-1, 0, +1\}$	[1, 3]
$\phi : x \mapsto x^3$	Frobenius automorphism in GF(9)	this doc.
$\varphi : N \mapsto N \bmod 3$	Ring homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$	this doc.
<i>FFGFT quantities</i>		
ξ	$= 4/30\,000$, geometric base parameter	[9]
Δw	Winding quantum (triality), $\Delta w \in \{0, +1, -1\}$	[10]
$T^4/\mathbb{Z}_3$	Compact 4-torus with $\mathbb{Z}_3$ orbifold	[9]
$k \leftrightarrow -k$	Sector pairing on $T^4/\mathbb{Z}_3$	[10]
<i>Matzke GALG</i>		
$V_s = V^\dagger$	Vacuum conjugation (adjunction)	[11]
$T_k^3 = \mathbf{1}$	Trine relation (tripotent condition)	[11]
$N = \sum q_k^\dagger q_k$	Number operator, $\sigma(N) = \{0, 1, 2, 3\}$	[4, 1]
<i>CKM notation</i>		
$\lambda \approx 0.225$	Wolfenstein parameter (Cabibbo angle)	[8]
$C_2 = 4/3$	Casimir value $SU(3)_{\text{fund}}$	Lie algebra
$ V_{ub} $	CKM element: $u \rightarrow b$ transition	[8]

Symmetric $\mathbb{Z}_3\mathbb{C}$: Elements $a + bi$ with $a, b \in \{-1, 0, +1\}$ use the symmetric representation of GF(3) around zero. $+1 + 1 = -1$ (ternary inline carry) and $-1 - 1 = +1$ (ternary inline borrow). The generator $(1 + i)^n$ has order 8.

.1 Motivation: Characteristic 3 and FFGFT

In characteristic 3, $(1 + N)^3 = 1$ for nilpotent N with $N^2 = 0$ holds *algebraically exactly* — because $\binom{3}{k} \equiv 0 \pmod{3}$ for $0 < k < 3$. The machine-epsilon result $|T_k^3 - \mathbf{1}| < 6.5 \times 10^{-16}$ from Doc. 324 is not an approximation; it does not, however, follow from characteristic 3: Doc. 324 computes over $\mathbb{C}$ with $T_k = \mathbf{1} + (\omega - 1)P$, $P^2 = P$, and $T_k^3 = \mathbf{1} + (\omega^3 - 1)P = \mathbf{1}$ holds exactly because $\omega^3 = 1$ **[B]**. Over $\mathbb{C}$ one has $(1 + N)^3 = 1 + 3N \neq 1$, and $(\omega - 1)P$ is not nilpotent; the reason is $\omega^3 = 1$ together with $P^2 = P$.

.2 Five structural questions — updated status

Question 1: Number operator N and FFGFT winding quantum Δw

Previously open **[S]; now partially closed.**

Furey's number operator $N = \sum_j q_j^\dagger q_j$ has spectrum $\{0, 1, 1, 1, 2, 2, 2, 3\}$ on the eight basis states of one generation. Reduction mod 3 gives:

$$N \bmod 3 = \{0, +1, +1, +1, -1, -1, -1, 0\} \quad (1)$$

with particle classification:

$$\begin{aligned} N \bmod 3 = 0 & : \text{ colour-neutral (neutrino, positron)} \\ N \bmod 3 = 1 & : \text{ anti-down quarks (3 colours)} \\ N \bmod 3 = 2 & : \text{ up quarks (3 colours)} \end{aligned}$$

This is *exactly* the $\mathbb{Z}_3$ class decomposition by the winding quantum $\Delta w \in \{0, +1, -1\}$ (triality, see below).

Ring homomorphism **[B]:**

$$\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}, \quad \varphi(N) = N \bmod 3 = \Delta w \quad (2)$$

is the canonical quotient; $\varphi(N + M) = \varphi(N) + \varphi(M) \bmod 3$ is a proved algebraic property.

Status: **[B] (homomorphism), **[K]** (physical assignment)**

$\varphi(N) = N \bmod 3$ is a ring homomorphism **[B]**. The physical identification $\varphi(N) = \Delta w$ (Furey's number operator $\leftrightarrow$ FFGFT winding quantum) classifies the same threefold particle spectrum decomposition; the deeper algebraic reason for this agreement is **[K]**.

Triality and orbifold sectors. $\Delta w = N \bmod 3$ is an additive $\mathbb{Z}_3$ charge, the triality; the ring homomorphism φ is unaffected by this. It is, however, not the sector index k of the orbifold action τ of Doc. 321: the winding sum is invariant under τ , and every three-element orbit contains states of all three sectors $\mathcal{H}_0, \mathcal{H}_1, \mathcal{H}_2$ (Doc. 321). The two $\mathbb{Z}_3$ are compatible, since the triality is invariant under τ , but they are different quantities. Which of the two is the physical colour charge remains open; Docs. 346 to 357 use the triality. The Frobenius fixed points $\{0, \pm 1\}$ in Section .3 are likewise values of the triality. Verification script: `pruef_341_zwei_z3.py`.

Question 2: Charge formula $Q = N/3$ and $\mathbb{Z}_3$ classification

Previously open **[S]**; now partially closed.

$Q = N/3 \in \mathbb{Q}$ cannot be embedded into $\text{GF}(3)$ (3 is a zero divisor in $\mathbb{Z}/9\mathbb{Z}$, no inverse). However an injection exists:

$$\iota : \{0, \frac{1}{3}, \frac{2}{3}, 1\} \hookrightarrow \mathbb{Z}/3\mathbb{Z} \times \{0, 1\}, \quad \iota(Q) = (3Q \bmod 3, [Q]) \quad (3)$$

with values:

Q	$\iota(Q)$	Particle
0	(0, 0)	Neutrino
1/3	(1, 0)	Anti-Down
2/3	(2, 0)	Up
1	(0, 1)	Positron

The $\mathbb{Z}/3\mathbb{Z}$ component of $\iota(Q)$ is exactly $N \bmod 3$.

Status: [B] (classification), [K] (Q as $\mathbb{Q}$ number)

$N \bmod 3$ fully classifies $Q = N/3$ **[B]**. Q itself (as rational number 1/3, 2/3) requires $\mathbb{Q}$ and is not a $\text{GF}(3)$ object **[K]**.

Question 3: Sector pairing $k \leftrightarrow -k$ and Frobenius automorphism

Main result; was **[S]**; now **[B]**.

The Frobenius automorphism in $\text{GF}(9)$:

$$\phi : x \mapsto x^3, \quad \phi(a + bi) = a - bi \quad [\text{field conjugation}] \quad (4)$$

operates simultaneously on three levels:

- **GF(9) field:** $a + bi \mapsto a - bi$ (conjugation)
- **FFGFT sector:** $k \mapsto -k \bmod 3$ ($\mathbb{Z}_3$ inversion)
- **Matzke vacuum:** $V \mapsto V_s = V^\dagger$ (matrix adjunction)
- `pruef_326_gf9_orbit_masse_weinberg.py`: Fixed points **[B]**, norm hierarchy **[B]**, Weinberg topology **[K]**
- `pruef_328_gluonen_tripotente_gf27_noether.py`: gluons **[B]**, tripotents **[B]**, $\text{GF}(27)$ **[K]**, Noether **[B]**
- `pruef_329_ckm_xi_gf27.py`: $\lambda = \xi^{1/6}$, $|V_{ub}| = \sqrt{\xi}/3$ **[K]**

Numerically verified: $x^3 = \bar{x}$ for all 8 non-zero $\text{GF}(9)$ elements **[B]**.

Status [B]

All three operations are algebraically isomorphic realisations of the Frobenius automorphism $x \mapsto x^3$.

Question 4: Casimir singularity and ξ in $\text{GF}(3)$

$\xi = (4/3)/10^4$: Casimir denominator $3 \equiv 0$ in $\text{GF}(3)$ is singular. $\xi = 4/30000 = 1/7500$ has no residue class mod 3 (denominator $\equiv 0 \bmod 3$, not invertible); $\equiv 1$ holds only for the numerator 4 of the notation $4/30000$ (or $\xi = (4/3)/10^4$ with $10^4 \equiv 1$, only the factor 1/3 is singular); since $3 \mid 7500$, $v_3(\xi) = -1$. The singularity marks the transition $\text{GF}(3) \rightarrow \mathbb{R}$ **[K]**.

Question 5: $\sqrt{2}$ in GF(9) and n_{thresh}

$i^2 = -1 \equiv 2 \pmod{3}$, so $\sqrt{2} = i \in \text{GF}(9)$ [B]. $n_{\text{thresh}} = 2\pi/(\sqrt{2} \ln 2) \approx 6.41$ (R93) thus has an algebraically exact GF(9) kernel.

.3 Three further structural results**Frobenius fixed points = triality values (Question 3, extended)**

The complete orbit structure under $\phi(x) = x^3$ in GF(9):

- **Fixed points** ($x^3 = x$): $\{0, +1, -1\} = \text{GF}(3)$ — exactly the values of the triality $\Delta_w \in \{0, +1, -1\}$ [B]
- **Non-fixed points**: 6 elements in 3 conjugate 2-cycles [B]: $(i \leftrightarrow 2i)$, $(1+i \leftrightarrow 1+2i)$, $(2+i \leftrightarrow 2+2i)$ — exactly the FFGFT pairs $(k, -k)$ for $k \neq 0$.

Stable states under Frobenius are the triality values; unstable states are the conjugate pairs.

Norm hierarchy GF(3) $\rightarrow$ GF(9) $\rightarrow \mathbb{R}(\xi)$ for masses

In GF(3), $1^2 \equiv 2^2 \equiv 1 \pmod{3}$ — muon and tau are *algebraically degenerate* as squares [B]. The GF(3) structure distinguishes only:

$$n^2 \in \{0, 1\} \pmod{3} : \quad 0 \leftrightarrow \text{massless}, \quad 1 \leftrightarrow \text{massive (generations 2+3 degenerate)} \quad (5)$$

In GF(9), the norm $\|x\| = (a^2 + b^2) \pmod{3}$ gives two classes [B]:

$$\begin{aligned} \|x\| = 1 & : \{+i, -i, +1, -1\} \quad \text{— purely real/imaginary (light)} \\ \|x\| = -1 & : \{+1+i, +1-i, -1+i, -1-i\} \quad \text{— mixed (heavy)} \end{aligned}$$

The full mass hierarchy requires ξ [K]: ξ breaks the GF(3) degeneracy and gives $m_\mu/m_e \approx 207$, $m_\tau/m_\mu \approx 17$ as continuous numbers.

Status: [B] (degeneracy), [K] (mass values)

Hierarchy: GF(3) gives whether massive/massless; GF(9) gives light/heavy; $\xi \in \mathbb{R}$ gives exact mass ratios. The singularity $C_2 = 4/3$ in GF(3) (Question 4) marks exactly the transition GF(3) $\rightarrow \mathbb{R}$.

Weinberg angle: GF(9) gives topology, ξ gives value

$\text{Gal}(\text{GF}(9)/\text{GF}(3)) = \mathbb{Z}_2$ explains *why* there are two gauge groups $\text{SU}(2)_L$ and $\text{U}(1)_Y$ — a degree-2 extension gives exactly two components [K].

The obvious GF(9) quotient $|\text{GF}(3)^*|/|\text{GF}(9)^*| = 2/8 = 0.250$ is not the numerical value of $\sin^2 \theta_{W_i}$; it lies 8 % above the $\overline{\text{MS}}$ value 0.231. The on-shell Weinberg angle $1 - M_W^2/M_Z^2$ is matched within 0.4 % by the icosahedral matrix element $|\langle v_0 | R_5 | v_1 \rangle|^2 = 2/9$ (Doc. 372 §3.1); for pole masses from $\tilde{T} \cdot m = 1$ the on-shell scheme is the natural one. The $\overline{\text{MS}}$ value is described by the scale flow from ξ (Doc. 323). The GF(9) statement of this section — a degree-2 extension explains the two gauge groups — is unaffected by the choice of numerical value.

Status [K]

GF(9) provides the algebraic justification for two gauge groups (degree-2 Galois extension). The numerical value follows in the on-shell scheme from the A_5 matrix element $2/9$ (Doc. 372), in the $\overline{\text{MS}}$ scheme from the ξ scale flow (Doc. 323).

.4 Status overview**Table 1:** Algebraic bridge questions FFGFT $\leftrightarrow$ GF(9) — updated status

#	Question	Result	Status
1a	Ring homo $\varphi(N) = N \bmod 3$	canonical, proved	[B]
1b	$\varphi(N) = \Delta_w$ physical	same threefold split	[K]
2a	$N \bmod 3$ classifies Q	injection, proved	[B]
2b	$Q = N/3$ as GF(3) object	impossible ($3 \equiv 0$)	[K]
3	$k \leftrightarrow -k$ vs. $V_s = V^\dagger$	Frobenius $x \mapsto x^3$	[B]
4	$C_2 = 4/3$ and ξ in GF(3)	singularity = GF(3) $\rightarrow \mathbb{R}$	[K]
5	$\sqrt{2}$ in GF(9), n_{thresh}	$i^2 = 2$, kernel exact	[B]
P3b	Fixed pts. = triality values, $(k, -k)$ pairs	orbit structure	[B]
P4b	$1^2 = 2^2 = 1$ in GF(3)	Gen. 2+3 degenerate	[B]
P4c	Norm $\in \{1, 2\}$ in GF(9)	light/heavy	[B]
P5b	$\text{Gal}(\text{GF}(9)/\text{GF}(3)) = \mathbb{Z}_2$	two gauge groups; value $2/9$ (Doc. 372)	[K]

.5 Five algebraic computations (script 328)**Gluon representations from bilaterals**

With the normalised number operator $N = \sum_k q_k^\dagger q_k$ (Furey normalisation $q_k = a_{p,k}/2$), the spectrum is $\sigma(N) = \{0, 1, 2, 3\}$ **[B]** — exactly Furey's result.

The *bilaterals* $q_i q_j^\dagger$ yield gluon representations:

- 6 off-diagonal $(q_i q_j^\dagger, i \neq j)$: traceless, rank 2 **[B]**
- 2 diagonal from $(N_i - N_j)/\sqrt{2}$: SU(3) Cartan generators **[B]**

The 8 matrices are linearly independent (rank 8) **[B]** — 8 gluon representations matching the 8 generators of SU(3).

Tripotent classification

All tripotents T with $T^3 = T$ in the 8-dimensional representation:

- $T_k = P_{p,k} - P_{n,k}$: eigenvalues $\{+1, -1\}$ in sym. $\mathbb{Z}_3\mathbb{C}$ **[B]**
- $P_{p,k}, P_{n,k}$ themselves: tripotents with eigenvalues $\{0, +1\}$ **[B]**
- 6 tripotent products $T_i \cdot T_j$ ($i \neq j$) **[B]**

Generator comparison: FFGFT trine vs. Matzke's $(1 + 1j)^n$

FFGFT trine: $T_k^3 = \mathbf{1}$ to 6.5×10^{-16} **[B]** (algebraically exact over $\mathbb{C}$ because $\omega^3 = 1$, $P^2 = P$, Doc. 324; not from characteristic 3, see section "Motivation"). Matzke's $(1 + 1j)$ analogue in

8×8: $D^8 = 256 \cdot \mathbf{1}$ (scaling factor 256 from the matrix representation). In sym. $\mathbb{Z}_3\mathbb{C}$: $(1+i)^8 = 1$ (order 8) **[B]**.

GF(27) and three generations

GF(27) = GF(3)[x]/($x^3 + 2x + 1$) (irreducible, also $x^3 + 2x + 2$) **[B]**.

Orbit structure of Frobenius $x \mapsto x^3$ in GF(27): 3 fixed points = GF(3), $(27 - 3)/3 = 8$ three-cycles **[B]**.

Status [K]

8 three-cycles in GF(27) correspond to 8 quark-colour states in 3 generations. The Galois group $\text{Gal}(\text{GF}(27)/\text{GF}(3)) = \mathbb{Z}_3$ may describe generation mixing (CKM matrix) **[S]**.

Noether conserved charge of the tripotent symmetry

The tripotent condition $X^3 = X$ is a $\mathbb{Z}_3$ -affine symmetry. By Noether's theorem, every symmetry generates a conserved charge.

The corresponding Noether charge is **[B]**:

$$Q_{\mathbb{Z}_3} = N \bmod 3 = \Delta_w \quad (6)$$

This is identical to the ring homomorphism φ from §2.1, i.e. the triality. It is compatible with the $\mathbb{Z}_3$ orbifold action τ on $T^4/\mathbb{Z}_3$ (invariant under τ), but it is not its sector index (see Question 1).

Status [B]

Noether charge of the tripotent symmetry ($X^3 = X$) = winding quantum $\Delta_w = N \bmod 3$ (triality). It is compatible with the $\mathbb{Z}_3$ orbifold action but a different quantity from its sector index; which of the two is the physical colour charge is open.

.6 CKM Wolfenstein hierarchy from ξ and GF(27) (script 329)

From the degrees of GF(9) and GF(27) over GF(3) and ξ , three precise relations emerge for the CKM Wolfenstein parameters.

$$\lambda = \xi^{1/6}$$

GF(9) and GF(27) have degrees $[\text{GF}(9) : \text{GF}(3)] = 2$ and $[\text{GF}(27) : \text{GF}(3)] = 3$ over GF(3). There is no tower $\text{GF}(3) \subset \text{GF}(9) \subset \text{GF}(27)$: $\text{GF}(3^d) \subset \text{GF}(3^n)$ holds only for $d \mid n$, and $2 \nmid 3$. Degree 6 is reached only by GF(729): $2 \cdot 3 = \text{lcm}(2, 3) = 6$ is the degree of the compositum $\text{GF}(9) \cdot \text{GF}(27) = \text{GF}(729)$ over GF(3). The exponent $1/6$ is therefore a motivated stipulation (a justification via GF(729) is outstanding); the numerical value $\xi^{1/6} = 0.22602$ (0.46 % deviation) is not affected **[K]**:

$$\lambda = \xi^{1/6} = \xi^{1/(2 \cdot 3)} = 0.2260 \quad (\text{exp.: } \lambda = 0.2250, \text{ dev. } 0.5 \%) \quad (7)$$

The Wolfenstein hierarchy is the ξ hierarchy in steps of $1/6$: $|V_{us}| \sim \xi^{1/6}$, $|V_{cb}| \sim \xi^{1/3}$, $|V_{ub}| \sim \xi^{1/2}$.

$$|V_{ub}| = \sqrt{\xi} \cdot (C_2 - 1) = \sqrt{\xi}/3$$

With $C_2 = 4/3$ (Casimir value $SU(3)_{\text{fund}}$), $C_2 - 1 = 1/3$ **[B]**, and:

$$|V_{ub}| = \sqrt{\xi} \cdot (C_2 - 1) = \frac{\sqrt{\xi}}{3} = 3.85 \times 10^{-3} \quad (\text{PDG: } 3.82 \times 10^{-3}, \text{ dev. } 0.8\%) \quad \textbf{[K]} \quad (8)$$

This relation connects the GF(9) extension ($\sqrt{\xi}$) with the $SU(3)$ Casimir value.

Consistency: $A \cdot |\vec{\rho}| \approx 1/3$

From $\lambda = \xi^{1/6}$ and $|V_{ub}| = \sqrt{\xi}/3$ it follows necessarily **[K]**:

$$A\sqrt{\rho^2 + \eta^2} = \frac{|V_{ub}|}{\lambda^3} = \frac{\sqrt{\xi}/3}{\xi^{1/2}} = \frac{1}{3} = C_2 - 1 \quad (9)$$

Numerically: $A\sqrt{\rho^2 + \eta^2} = 0.316$ vs. $1/3 = 0.333$ (5% dev.).

Status [K]

CKM Wolfenstein hierarchy from the field degrees of GF(9), GF(27) and ξ :

$$\lambda = \xi^{1/(\deg(\text{GF}9) \cdot \deg(\text{GF}27))} = \xi^{1/[\text{GF}(729) : \text{GF}(3)]} = \xi^{1/6}$$

$$|V_{ub}| = \sqrt{\xi} \cdot (C_2 - 1) = \sqrt{\xi}/3$$

The exponent is a motivated stipulation (degree of the compositum, not of a tower up to GF(27)). Verification script: `pruef_329_ckm_xi_gf27.py` (16 assertions, all OK).

.7 ξ as a Galois Number

The mass-ratio identity from §.6 leads to a new result: $\xi = 4/30000$ is not a free constant but a Galois number **[K]**.

Core identity

The FFGFT formula (Doc. 006) and the Galois group orders from §.3 yield:

$$\frac{(r_\mu/r_e)^2}{\xi} = |\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)| = 8^2 \cdot 25 \cdot 27 = 43200 \quad (10)$$

These are not two computations of the same value but a single algebraic equation. From it ξ follows:

$$\xi = \frac{(r_\mu/r_e)^2}{43200} = \frac{(12/5)^2}{43200} = \frac{1}{7500} = \frac{4}{30000} \quad \textbf{[K]} \quad (11)$$

Winding numbers from $\text{GF}(3^n)$

The quantum numbers $r_g = n_{\phi,g}^2/n_{\theta,g}$ (Doc. 317) follow from a recursion with a Galois initial condition:

Winding number	Value	Galois origin
$n_{\theta,1}$	3	$\text{char}(\text{GF}(3^n))$
$n_{\phi,1}$	2	$ \text{GF}(3)^* $
$n_{\theta,2}$	5	recursion = $3 + 2$; largest or primitive prime factor of $ \text{GF}(81)^* $
$n_{\phi,2}$	4	$\varphi(5)$, Euler- φ of the prime factor
$n_{\theta,3}$	9	recursion = $5 + 4$
$n_{\phi,3}$	5	largest or primitive prime factor of $ \text{GF}(81)^* = 80$

Since $80 = 2^4 \cdot 5$, 5 is the largest and the only primitive (Zsigmondy) prime factor of $3^4 - 1$; it is new compared to the earlier stages only in this sense, since $|\text{GF}(27)^*| = 26$ already contributes 13.

All factors in the core identity are Galois orders: $8 = |\text{GF}(9)^*|$, $27 = |\text{GF}(27)|$, $5 =$ largest or primitive prime factor of $|\text{GF}(81)^*|$. Verification scripts: `pruef_330` (mass ratio), `pruef_331` (ξ derivation).

Verification scripts

- `pruef_324_z3c_trine_exakt.py`: Algebraic proof $T_k^3 = \mathbf{1}$ (over $\mathbb{C}$ from $\omega^3 = 1$, $P^2 = P$) **[B]**
- `pruef_325_gf9_ffgft_bruecke.py`: All five questions; Frobenius **[B]**; ring homo **[B]**; injection ι **[B]**

Note on AI assistance

This document was produced with AI-assisted support (Claude, Anthropic). All algebraic results were independently verified by Python verification scripts; the author bears sole responsibility for the content.

The methodological principles governing the use of AI in FFGFT documents are described in the A-series (in particular Docs. A001–A010): AI-generated content is only incorporated into the corpus after independent review and verification by scripts. External criticism and AI outputs are not adopted without scrutiny (Doc. 190, methodology note).

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